LSPU COURSE GUIDE

Course	System Integration and Architecture 2
Sem/AY	Second Semester/2025-2026

Sample Narrative Report Structure: SYSTEM INTEGRATION AND ARCHITECTURE 2

An ERP system a PROJECT MANAGEMENT for the course System Integration and Architecture 2

1. Introduction:

- **Project Overview:**

Briefly describe the project, including the client, purpose, and key goals of the new system.

- **Problem Statement:**

Clearly state the existing system's major issues and pain points that the new system aims to address.

- **Project Scope:**

Define the boundaries of the project, what functionalities are included, and any limitations.

2. System Analysis:

- **Current System Evaluation:**
    - Detailed description of the existing system's processes, functionalities, data flow, and key components.
    - Identify strengths, weaknesses, inefficiencies, and potential areas for improvement.
    - Include data gathered through interviews, observations, and document analysis.
- **User Requirements Gathering:**
    - Summarize user needs and expectations through requirements elicitation techniques like user interviews, surveys, and workshops.
    - Categorize requirements into functional (what the system should do) and non-functional (performance, security, usability) requirements.
- **Data Analysis and Modeling:**

- Present data flow diagrams (DFDs), entity-relationship diagrams (ERDs), and other visual models to represent the system's data relationships and processing logic.

3. System Design:

- **Proposed System Architecture:**
  - Describe the overall system architecture, including hardware, software components, network topology, and database design.
  - Justify the chosen technology stack with its benefits and limitations.
- **Functional Design:**
  - Detail the design of key system functionalities, user interfaces, and interactions with specific use cases and scenarios.
- **Data Design:**
  - Explain the data structure, data dictionary, and database schema to effectively store and retrieve system data.

4. Implementation:

- **Development Process:**
  - Outline the development methodology used (e.g., Agile, Waterfall) and key phases like coding, testing, and deployment.
  - Highlight any challenges encountered and solutions implemented.
- **System Integration:**
  - Describe how different system components were integrated and tested to ensure seamless functionality.

5. Testing and Evaluation:

- **Test Strategy:**
  - Explain the testing approach, including types of tests (unit, integration, system, acceptance) and testing criteria.
- **Test Results:**
  - Summarize test results, identifying any issues or defects found and their resolution process.
  - Include performance metrics like response time and throughput where relevant.

6. Conclusion and Recommendations:

- **Key Findings:**
  - Summarize the most significant insights gained from the system analysis and design process.
- **Recommendations:**
  - Provide actionable recommendations for future system enhancements, maintenance, and potential areas for further optimization.

Final Presentation May 19 – 26, 2025

System and Narrative

COURSE CALENDAR/SCHEDULE

Week	Date/Period	Topic/s:	Activities
1	Feb 3, 2025	0. **Course Primer**	1. Distribution of Course Guide 2. Discussion of the following: a. course syllabus b. classroom management c. LMS Orientation d. module orientation e. laboratory activities
2-4	February 13,2026	**I. Introduction.** • Project Overview: Project Overview: Briefly describe the project, including the client, purpose, and key goals of the new system. • Problem Statement: Clearly state the existing system's major issues and pain points that the new system aims to address. • Project Scope: Define the boundaries of the project, what functionalities are included, and any limitations.	Client/company profile 2. Acceptance letter from client 3. Data privacy disclosure agreement 4. Data gathering/ Client interview 5. Presentation of biographical sketch of the project team leader. 6. Oral presentation of proposed project
5-6	Feb 27, 2026	**II. System Analysis** • Current System Evaluation:	1. Group collaborative activity

		• Detailed description of the existing system's processes, functionalities, data flow, and key components. • Identify strengths, weaknesses, inefficiencies, and potential areas for improvement. • Include data gathered through interviews, observations, and document analysis. • **User Requirements Gathering:** • Summarize user need and expectations through requirements elicitation techniques like user interviews, surveys, and workshops. • Categorize requirements into functional (what the system should do) and non-functional (performance, security, usability) requirements. • Data Analysis and Modeling: Present data flow diagrams (DFDs), entity-relationship diagrams (ERDs), and other visual models to represent the system's data relationships and processing logic.	2. Oral presentation of project scope and system flow.
Week 7-8	March 6, 2026	**System Design** • Proposed System Architecture: • Describe the overall system architecture, including hardware, software components, network topology, and database design. • Justify the chosen technology stack with its benefits and limitations. Functional Design:	1. Group collaborative activity Oral presentation of system design Gmeet and onsite activity

		• Detail the design of key system functionalities, user interfaces, and interactions with specific use cases and scenarios. Data Design: • Explain the data structure, data dictionary, and database schema to effectively store and retrieve system data.	

Week 9-12	**March 13, 2026**	Implementation
		
		• **Development Process:**
		  • Outline the development methodology used (e.g., Agile, Waterfall) and key phases like coding, testing, and deployment.
		  • Highlight any challenges encountered and solutions implemented.
		• **System Integration:**
		  • Describe how different system components were integrated and tested to ensure seamless functionality.
			1 Group collaborative activity
			Oral presentation of system design
			Gmeet and onsite activity

Week 13-15	**April 3, 2026**	**Testing and Evaluation:** • **Test Strategy:** • Explain the testing approach, including types of tests (unit, integration, system,	1 Group collaborative activity Oral presentation of system design Gmeet and onsite activity

		acceptance) and testing criteria. • **Test Results:** • Summarize test results, identifying any issues or defects found and their resolution process. • Include performance metrics like response time and throughput where relevant.	Testing strategy Questionnaire System test to client/admin/user and IT Professionals
Week 16-17	**April 24 2026**	Conclusion and Recommendations: • Key Findings: • Summarize the most significant insights gained from the system analysis and design process. • Recommendations: • Provide actionable recommendations for future system enhancements, maintenance, and potential areas for further optimization.	Data analysis And interpretation
Week 18	**May 18-22**	**Final Project presentation** **Finals**	**Oral presentation**

Important Considerations:

- **Audience:**

Tailor the report language and level of technical detail to the intended audience (e.g., technical team, project stakeholders).

- **Visual Aids:**

Utilize diagrams, charts, and tables to effectively present complex information.

- **Clarity and Conciseness:**

Ensure the report is well-organized, easy to navigate, and avoids unnecessary jargon.

Remember to adapt this sample structure to fit the specific needs and context of your project.

Key takeaways

- ERP software manages financial activities, accounting, bookkeeping, project management, and supply chain operations.
- They create a single source of truth for employees and executives.
- Businesses should assess ERP providers' capabilities, implementation models, integration needs, and total cost of ownership before choosing a system.

I am your Faculty-in-Charge. There is more information about me in our course website in LMS. You may reach me at Faculty office.

Name of Faculty: **Myline V. Anquilo**

Laguna State Polytechnic University

Campus: LOS BANOS LAGUNA

Campus Address: Barangay Mayondon - Malinta, 4030 Los Baños, Laguna